



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 1

Candidate Name

Class

CHEMISTRY

Preliminary Examination
Paper 1 Multiple Choice

8872/01

30 August 2013
1 hour

Additional Materials: Data Booklet
 Optical Mark Sheet (OMS)

READ THESE INSTRUCTIONS FIRST

On the separate multiple choice OMS given, write your name, FIN/NRIC and class in the spaces provided.

Shade correctly your class and FIN/NRIC number.

Eg. If your NRIC is S9306660Z, shade **S9306660Z** for the item "index number".

There are **forty** questions in this paper. Answer **all** questions.

For each question there are four possible answers **A, B, C** and **D**.

Choose the one you consider correct and record your choice using a **soft pencil** on the separate OMS.

Each correct answer will score one mark.

A mark will not be deducted for a wrong answer.

You are advised to fill in the OMS as you go along; no additional time will be given for the transfer of answers once the examination has ended.

Any rough working should be done in this question paper.

Section A
Answer all questions

1	The percentage by mass of water in a hydrated manganese(II) chloride salt is 36.4%. What is the empirical formula of the hydrated salt?	
	A	$\text{MnCl}_2 \cdot 2\text{H}_2\text{O}$
	B	$\text{MnCl}_2 \cdot 3\text{H}_2\text{O}$
	C	$\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$
	D	$\text{MnCl}_2 \cdot 5\text{H}_2\text{O}$
Answer: C Let the empirical formula of the hydrated salt be $\text{MnCl}_2 \cdot x\text{H}_2\text{O}$ Hence, % by mass of water in $\text{MnCl}_2 \cdot x\text{H}_2\text{O} = (18.0x)(100)/(54.9 + 71 + 18.0x) = 36.4\%$ Solving x, x = 4		

2	In which one of the following reactions is there a reduction in the oxidation state of the underlined element?	
	A	$\underline{\text{N}}\text{H}_3 + \text{HCl} \rightarrow \underline{\text{N}}\text{H}_4\text{Cl}$
	B	$2\underline{\text{V}}^{3+} + \text{H}_2\text{O}_2 \rightarrow 2\underline{\text{V}}\text{O}^{2+} + 2\text{H}^+$
	C	$\text{H}_2\underline{\text{O}}_2 + 2\text{I}^- + 2\text{H}^+ \rightarrow 2\text{H}_2\underline{\text{O}} + \text{I}_2$
	D	$\text{Ca}\underline{\text{C}}\text{O}_3 \rightarrow \text{CaO} + \underline{\text{C}}\text{O}_2$
Answer: C A $\underline{\text{N}}\text{H}_3 + \text{HCl} \rightarrow \underline{\text{N}}\text{H}_4\text{Cl}$ (remains at -3) B $2\underline{\text{V}}^{3+} + \text{H}_2\text{O}_2 \rightarrow 2\underline{\text{V}}\text{O}^{2+} + 2\text{H}^+$ (increases from +3 to +4) C $\text{H}_2\underline{\text{O}}_2 + 2\text{I}^- + 2\text{H}^+ \rightarrow 2\text{H}_2\underline{\text{O}} + \text{I}_2$ (reduces from -1 to -2) D $\text{Ca}\underline{\text{C}}\text{O}_3 \rightarrow \text{CaO} + \underline{\text{C}}\text{O}_2$ (remains at -4)		

3	Which of the following particles has more electrons than protons, and more protons than neutrons? [$H = {}^1_1H$, $D = {}^2_1H$, $He = {}^4_2He$, $O = {}^{16}_8O$]	
	A	He^+
	B	D^-
	C	OD^-
	D	OH^-
Answer: D A He^+ (4 protons, 3 electrons, 2 neutrons) B D^- (1 protons, 2 electrons, 1 neutrons) C OD^- (8 + 1 protons, 8 + 1 + 1 electrons, 8 + 1 neutrons) D OH^- (8 + 1 protons, 8 + 1 + 1 electrons, 8 neutrons)		

4	Under the same conditions, which of the following ions would be deflected in a mass spectrometer to the same extent as ${}^{12}_6C^{2+}$?	
	(i) ${}^6_3Li^+$ (ii) ${}^{12}_6C^+$ (iii) ${}^{12}_7N^{2+}$ (iv) ${}^{13}_6C^{2+}$	
	A	(i), (ii), (iii) and (iv)
	B	(i) and (iii) only
	C	(ii) only
	D	None of the above.
Answer: B Angle of deflection $\propto e/m$ $e/m ({}^{12}_6C^{2+}) = 2/12 = 1/6$ $e/m ({}^6_3Li^+) = 1/6$ $e/m ({}^{12}_6C^+) = 1/12$ $e/m ({}^{12}_7N^{2+}) = 2/12 = 1/6$ $e/m ({}^{13}_6C^{2+}) = 2/13$		

5	Which of the following statements best explains why the boiling point of $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ is higher than that of $(\text{CH}_3)_3\text{N}$?	
	A	$\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ has a higher relative molecular mass than $(\text{CH}_3)_3\text{N}$.
	B	The $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ molecule has a larger surface area than the $(\text{CH}_3)_3\text{N}$ molecule.
	C	$\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ is polar but $(\text{CH}_3)_3\text{N}$ is not.
	D	There are hydrogen bonds between $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ molecules but not between $(\text{CH}_3)_3\text{N}$ molecules.
Answer: D		

6	Although F–F bond is the shortest among the halogens, it has the weakest bond strength. What could be the possible reason for the unusual phenomenon?	
	A	Fluorine has the highest electronegativity in Group VII.
	B	Fluorine has the smallest atomic radius in Group VII.
	C	Due to lone pair-lone pair repulsion between the 2 fluorine atoms.
	D	Due to the non-polar nature of the F–F bond.
Answer: C		

7	Orange dichromate, $\text{Cr}_2\text{O}_7^{2-}$ and yellow chromate, CrO_4^{2-} exist in equilibrium in aqueous solution. $\text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{O} \rightleftharpoons 2\text{CrO}_4^{2-} + 2\text{H}^+$ Which statement about this equilibrium is correct?	
	A	$\text{Cr}_2\text{O}_7^{2-}$ is reduced to CrO_4^{2-} .
	B	The solution appears to be yellow in NaOH (aq).
	C	K_c value increases with the addition of NaOH (aq).
	D	Lowering of pH will increase the concentration of CrO_4^{2-} .
Answer: B A: Both contains Cr in O.S. of +6 B: In NaOH, $[\text{H}^+]$ decreases, by LCP, position of equilibrium shift to the right resulting in an increase in $[\text{CrO}_4^{2-}]$ causing the solution to appear yellow. C: Value of K_c is temperature dependent only. D: Lowering pH results in increase $[\text{H}^+]$ by LCP, position of equilibrium shift left, decreasing $[\text{CrO}_4^{2-}]$.		

8	Each of the following equilibria is subjected to two changes which are carried out separately: (i) The pressure is reduced at constant temperature; (ii) The temperature is increased at constant pressure. For which equilibrium will both of these changes result in an increase in the proportion of products?																	
	A	$\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g});$	$\Delta H = +53 \text{ kJ mol}^{-1}$															
	B	$4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightleftharpoons 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{g});$	$\Delta H = -950 \text{ kJ mol}^{-1}$															
	C	$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g});$	$\Delta H = -92 \text{ kJ mol}^{-1}$															
	D	$\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}(\text{g});$	$\Delta H = +57 \text{ kJ mol}^{-1}$															
Answer: D (for both (i) and (ii) position of equilibrium shift right)																		
<table><tr><th>Option</th><th>Reduce P on P.O.E.</th><th>Increase T on P.O.E.</th></tr><tr><td>A</td><td>No change</td><td>Right</td></tr><tr><td>B</td><td>Right</td><td>Left</td></tr><tr><td>C</td><td>Left</td><td>Left</td></tr><tr><td>D</td><td>Right</td><td>Right</td></tr></table>				Option	Reduce P on P.O.E.	Increase T on P.O.E.	A	No change	Right	B	Right	Left	C	Left	Left	D	Right	Right
Option	Reduce P on P.O.E.	Increase T on P.O.E.																
A	No change	Right																
B	Right	Left																
C	Left	Left																
D	Right	Right																

9	Nitrogen dioxide decomposes on heating according to the following equation $2\text{NO}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g}) + \text{O}_2(\text{g})$ When 4 moles of nitrogen dioxide was put into a 1 dm ³ container and heated, the equilibrium mixture contained 0.8 moles of oxygen. What is the numerical value of the equilibrium constant, K_c , at the temperature of the experiment?	
	A	$\frac{1.6 \times 0.8}{2.4^2}$
	B	$\frac{1.6^2 \times 0.8}{2.4^2}$
	C	$\frac{0.8 \times 0.8}{2.4}$
	D	$\frac{0.8^2 \times 0.8}{4^2}$
	Answer: B	

10	Values for the ionic product of water, K_w , at two different temperatures are given below.							
	<table><tr><td>temperature / °C</td><td>$K_w / \text{mol}^2 \text{dm}^{-6}$</td></tr><tr><td>25</td><td>1.00×10^{-14}</td></tr><tr><td>30</td><td>1.44×10^{-14}</td></tr></table>		temperature / °C	$K_w / \text{mol}^2 \text{dm}^{-6}$	25	1.00×10^{-14}	30	1.44×10^{-14}
temperature / °C	$K_w / \text{mol}^2 \text{dm}^{-6}$							
25	1.00×10^{-14}							
30	1.44×10^{-14}							
	Which is correct for pure water at 30 °C?							
	A	pH < 7						
	B	$[\text{H}^+] < [\text{OH}^-]$						
	C	$[\text{H}^+] = 1.44 \times 10^{-7} \text{ mol dm}^{-3}$						
	D	dissociation of water is an exothermic process						
	Answer: A A: $[\text{H}^+] = \sqrt{1.44 \times 10^{-14}} = 1.2 \times 10^{-7} \text{ mol dm}^{-3}$; pH = -lg (1.2 x 10⁻⁷) = 6.92 B: $[\text{H}^+] = [\text{OH}^-]$ C: $[\text{H}^+] = \sqrt{1.44 \times 10^{-14}} = 1.2 \times 10^{-7} \text{ mol dm}^{-3}$ D: K_w value increases with increased in temperature, hence dissociation of water is an endothermic process.							

11	Lead is the final product formed by a series of changes in which the rate-determining stage is the radioactive decay of uranium-238. This decay is a first-order reaction with half-life of 4.5×10^9 years. What would be the age of a rock sample originally lead-free, in which molar ratio of uranium to lead is now 0.143?	
	A	1.5×10^9 years
	B	4.5×10^9 years
	C	12.6×10^9 years
	D	13.5×10^9 years
	Answer D 100% U → 50% U → 25% U → <u>12.5% U</u> → 6.25% U 0% Pb → 50% Pb → 75% Pb → <u>87.5% Pb</u> → 93.75% When 12.5% U : 87.5% Pb → molar ratio is 0.143. It has went through 3 half lives $3 \times 4.5 \times 10^9 = 1.35 \times 10^{10}$	

12	A reaction involving reactant R is found out to be first order. Which of the following graphs best describes the decay of reactant R ?
	A B C D

Answer: B

A is completely wrong.

C is more for zero order.

B and D are close match.

B is a better graph as it shows the linear relationship of Rate = $k[R]$

D, however, can represent a first order or second order graph. If B is not present then D will make a better choice, which is not a case for this question.

13	The magnitude of enthalpy change of neutralisation of aqueous sodium hydroxide by hydrochloric acid is 57.2 kJ mol^{-1}, but the magnitude of enthalpy change of neutralisation of aqueous sodium hydroxide by aqueous ethanoic acid is 55.2 kJ mol^{-1}. Which statement best explains the numerical difference in these values?								
	<table border="1"> <tr> <td>A</td><td>The process $\text{CH}_3\text{CO}_2\text{H (aq)} \rightarrow \text{CH}_3\text{CO}_2^- \text{ (aq)} + \text{H}^+ \text{ (aq)}$ is exothermic</td></tr> <tr> <td>B</td><td>The process $\text{CH}_3\text{CO}_2\text{H (aq)} \rightarrow \text{CH}_3\text{CO}_2^- \text{ (aq)} + \text{H}^+ \text{ (aq)}$ is endothermic</td></tr> <tr> <td>C</td><td>Aqueous ethanoic acid is a weak acid and has fewer hydrogen ions than hydrochloric acid of the same concentration.</td></tr> <tr> <td>D</td><td>A smaller volume of aqueous hydrochloric acid is required for the neutralisation compared to that of ethanoic acid of the same concentration.</td></tr> </table>	A	The process $\text{CH}_3\text{CO}_2\text{H (aq)} \rightarrow \text{CH}_3\text{CO}_2^- \text{ (aq)} + \text{H}^+ \text{ (aq)}$ is exothermic	B	The process $\text{CH}_3\text{CO}_2\text{H (aq)} \rightarrow \text{CH}_3\text{CO}_2^- \text{ (aq)} + \text{H}^+ \text{ (aq)}$ is endothermic	C	Aqueous ethanoic acid is a weak acid and has fewer hydrogen ions than hydrochloric acid of the same concentration.	D	A smaller volume of aqueous hydrochloric acid is required for the neutralisation compared to that of ethanoic acid of the same concentration.
A	The process $\text{CH}_3\text{CO}_2\text{H (aq)} \rightarrow \text{CH}_3\text{CO}_2^- \text{ (aq)} + \text{H}^+ \text{ (aq)}$ is exothermic								
B	The process $\text{CH}_3\text{CO}_2\text{H (aq)} \rightarrow \text{CH}_3\text{CO}_2^- \text{ (aq)} + \text{H}^+ \text{ (aq)}$ is endothermic								
C	Aqueous ethanoic acid is a weak acid and has fewer hydrogen ions than hydrochloric acid of the same concentration.								
D	A smaller volume of aqueous hydrochloric acid is required for the neutralisation compared to that of ethanoic acid of the same concentration.								

Answer: B

The process $\text{CH}_3\text{CO}_2\text{H (aq)} \rightarrow \text{CH}_3\text{CO}_2^- \text{ (aq)} + \text{H}^+ \text{ (aq)}$ is endothermic means energy need ti be absorbed to cause the dissociation of the H^+ from the acid resulting in a smaller magnitude for enthalpy change of neutralisation.

C is not a wrong statement but it **does not fully explained** the lower magnitude of enthalpy change of neutralisation

D does not take into account energetics, it is related to volumetric analysis.

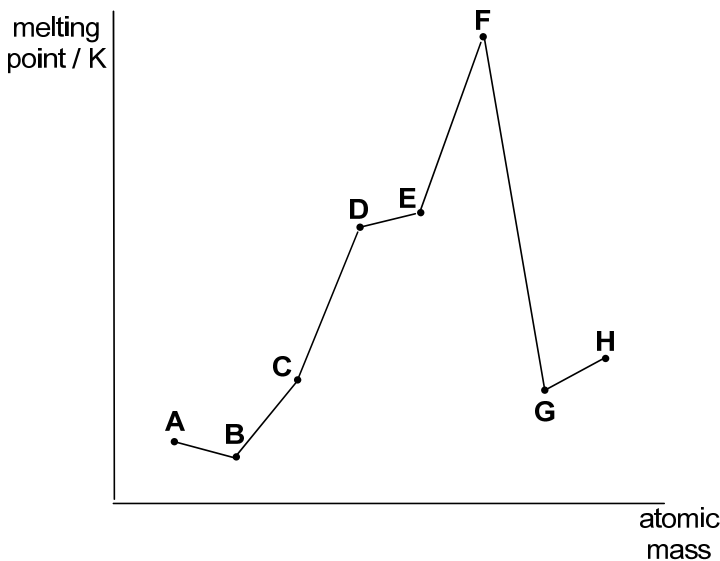
14	SiCl ₄ is known to hydrolyse readily in water but CCl ₄ is inert to water. Which statement best explains this observation?	
	A	CCl ₄ has a giant molecular structure, large amount of energy is required to overcome the strong covalent bond to allow solvation.
	B	There are vacant orbitals to accommodate additional electrons in SiCl ₄ but not in CCl ₄ .
	C	CCl ₄ has weaker van der Waals' forces of attraction which cannot displace the stronger hydrogen bonding in water.
	D	Si ⁴⁺ has high charge density to polarise the large chlorine electron cloud.

Answer: B

A is wrong as CCl₄ is not a giant molecular structure.

C is wrong as this statement explain physical property of hydration rather than relating to the inertness of CCl₄

D is wrong as SiCl₄ is not ionic and thus concept of high density is not applicable.

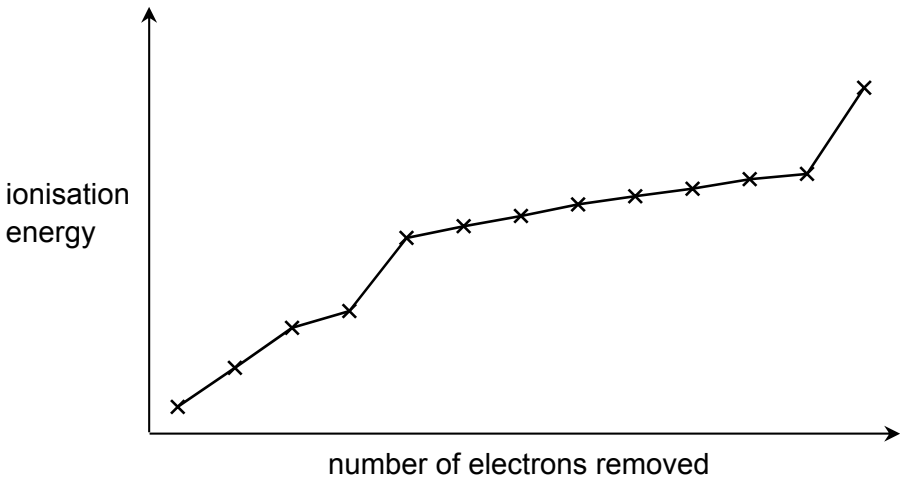
15	<i>Use of the Data Booklet is relevant to this question.</i> The graph below shows the variation in the melting points for eight consecutive elements in the Periodic Table, all with atomic number below 20.  What statement is correct?	
	A	Element F conducts electricity at room temperature.

	B	Element D burns with a brilliant yellow flame.
	C	Element H does not react with air at room temperature.
	D	Element C is a gas which is chemically inert at room temperature.

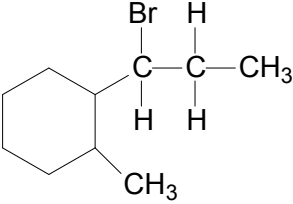
Ans: **C**

A is F, B is Ne, C is Na, D is Mg, E is Al, F is Si, G is P, H is S

S burns in oxygen to form SO_2 .

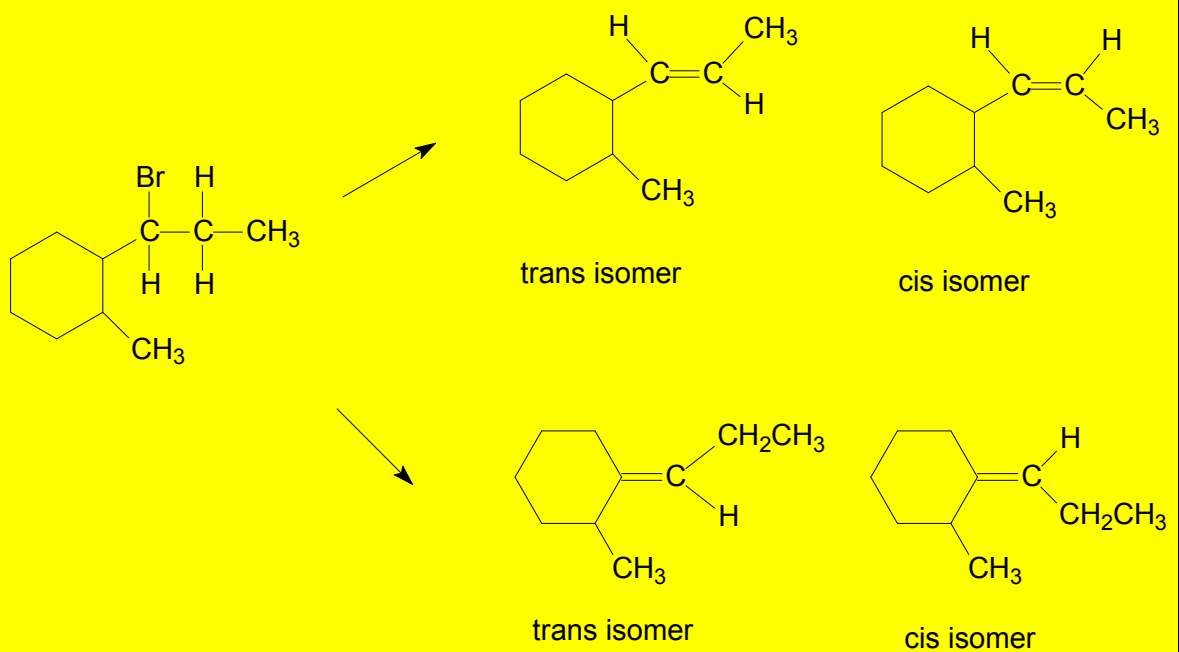
16	The graph below shows the first thirteen ionisation energies for element J.  What can be deduced from the graph about element J?	
	A	It is a d-block element.
	B	It is in Group IV of the Periodic Table.
	C	It has one electron in its outermost shell.
	D	It is in the third period (Na to Ar) of the Periodic Table.

Answer: **B**

17	How many different alkenes, including geometrical isomers, could be produced when  reacts with hot ethanolic NaOH?				
----	--	--	--	--	--

A	2	B	3	C	4	D	5
----------	---	----------	---	----------	---	----------	---

Answer: **C**



18	2,2-dimethylbutane undergoes free radical substitution with chlorine under ultraviolet light. How many different mono-substituted chlorinated products are possible?							
	A	2	B	3	C	4	D	5

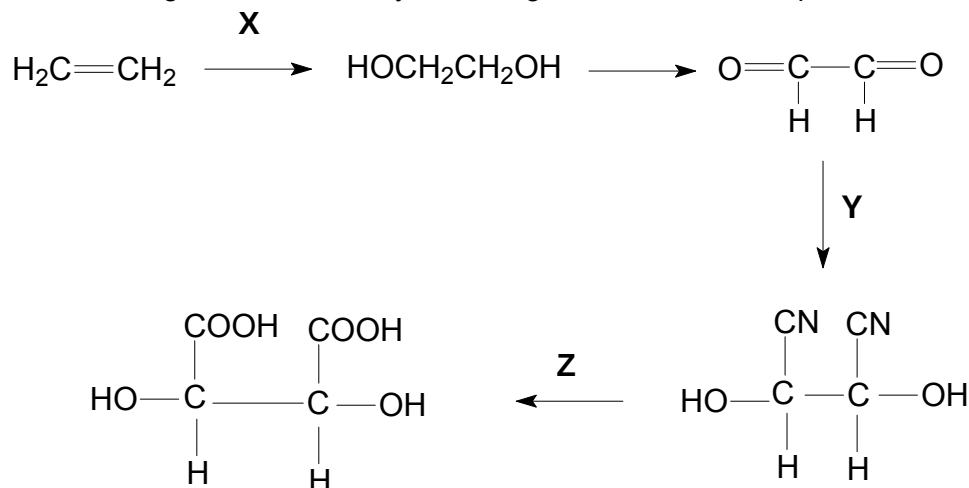
Answer: **B**

The diagram illustrates the free radical chlorination of 2,2-dimethylbutane. The reactant molecule is shown with its four types of hydrogen atoms highlighted: primary (green), secondary (red), and two types of tertiary (blue). Three arrows point to the resulting products:

- 1-chloro-2,2-dimethylbutane (from primary H)
- 2-chloro-2,2-dimethylbutane (from secondary H)
- 2-chloro-3,3-dimethylbutane (from tertiary H)

19	Chlorofluorocarbons (CFCs) have been widely used in aerosol sprays, refrigerators and in making foamed plastics, but are now known to destroy ozone in the upper atmosphere.	
	Which of the following will not destroy ozone, and therefore can be used as a safer replacement for CFCs?	
	A	$\text{CH}_2\text{FCH}_2\text{CH}_2\text{CH}_3$
	B	CCl_3CBr_3
	C	CHBr_3
	D	$\text{CHCl}_3\text{CFCl}_2$
Answer: A C-F bond is strong and it is difficult to break the C-F bond to form radicals.		

20 The following is a method of synthesising tartaric acid, a compound found in wine.



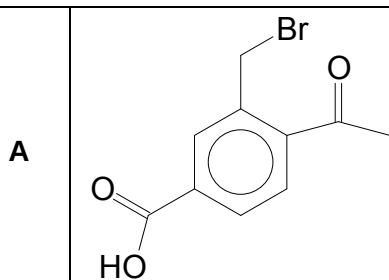
Which set of reagents and conditions can be used for the synthesis?

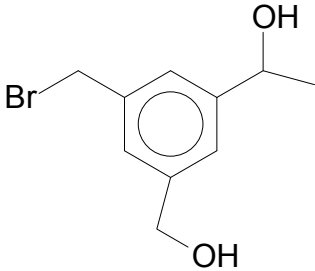
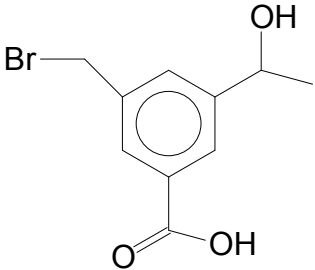
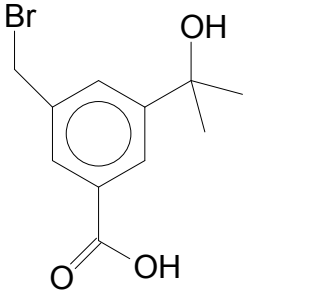
	Stage X	Stage Y	Stage Z
A	H ₂ O (g), H ₃ PO ₄ (aq)	Cold HCN, NaOH (aq)	Hot K ₂ Cr ₂ O ₇ , H ₂ SO ₄ (aq)
B	Cold KMnO ₄ , H ₂ SO ₄ (aq)	Ethanolic KCN, heat	HCl (aq), heat
C	H ₂ O (g), H ₃ PO ₄ (aq)	Ethanolic KCN, heat	Hot K ₂ Cr ₂ O ₇ , H ₂ SO ₄ (aq)
D	Cold KMnO ₄ , NaOH (aq)	Cold HCN, NaOH (aq)	HCl (aq), heat
Answer: D			

21 Data about the reactions of compound **K** are given in the table.

Reaction	Reagent	Result
1	Hot acidified potassium dichromate, followed by 2,4-DNPH	Orange solution turns green and orange precipitate seen after addition of 2,4-DNPH.
2	Warm alkaline aqueous iodine	Yellow precipitate observed.
3	Aqueous sodium carbonate	Effervescence seen.
4	AgNO ₃ (aq), warm	Cream solid formed.

Which of the following compounds is **K**?



	B	
	C	
	D	

Answer: **C**

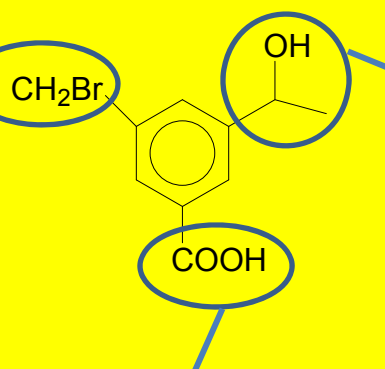
Option A is incorrect as the compound will not reduce orange potassium dichromate to green.

Option B is incorrect as there is a lack of a carboxylic acid functional group to react with aqueous sodium carbonate.

Option D is incorrect as Br attached directly to benzene could have no reaction with AgNO_3 .

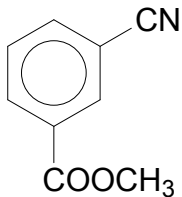
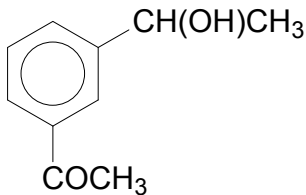
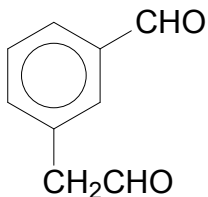
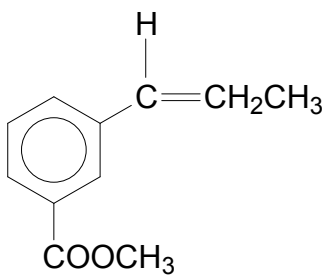
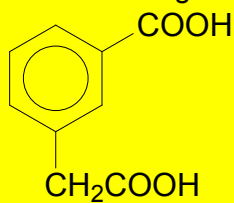
Option C is correct as

- Produce cream ppt of AgBr with silver nitrate.



- Able to reduce orange dichromate to green Cr^{3+} . Ketone would be formed and thus orange ppt with 2,4-DNPH
- Has the special structure to produce yellow ppt with aqueous alkaline iodine.

- Able to produce CO_2 when reacted with aq sodium carbonate.

22	Which of the following reactions will not produce 1,3-dibenzoic acid?	
	A	 <chem>COC(=O)c1ccc(C#N)cc1</chem> + aqueous H_2SO_4 , reflux
	B	 <chem>CC(=O)c1ccc(C(C)O)cc1</chem> + alkaline aqueous iodine followed by acidification
	C	 <chem>O=Cc1ccc(C=O)cc1</chem> + Tollens' reagent followed by acidification
	D	 <chem>CCOC(=O)c1ccc(C=C)cc1</chem> + hot acidified KMnO_4
Answer: C Tollens' reagent can react with both aliphatic and aromatic aldehyde. But compound in choice C will give the following product:  <chem>OC(=O)c1ccc(C(=O)O)cc1</chem>		

23	Citric acid can be found in lemon juice and orange juice and it causes the sharp unique taste. $ \begin{array}{c} \text{CH}_2\text{---COOH} \\ \\ \text{HO---C---COOH} \\ \\ \text{CH}_2\text{---COOH} \end{array} $ Which of the following reacts completely with 1 mole of citric acid?
	A 3 moles of PCl_5
	B 3 moles of NaHCO_3
	C 4 moles of H_2SO_4
	D 4 moles of KOH
Answer : B	

24	Arrange the following molecules in terms of increasing $\text{p}K_a$ values. Compound L $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$ Compound M $\text{CH}_2(\text{Br})\text{CH}_2\text{CH}_2\text{COOH}$ Compound N $\text{CH}_3\text{CH}_2\text{CH}(\text{Br})\text{COOH}$ Compound P $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{COOH}$
	A $\text{M} < \text{N} < \text{L} < \text{P}$
	B $\text{M} < \text{N} < \text{P} < \text{L}$
	C $\text{N} < \text{M} < \text{L} < \text{P}$
	D $\text{N} < \text{P} < \text{L} < \text{M}$
Answer: C Increasing $\text{p}K_a \rightarrow$ Decreasing acidity Factors that decrease the acidity: • Electron donating group will intensify the negative charge on carboxylate ion, destabilising it, thus lowering acidity • The closer the electron donating to the COO^-, the more destabilised the carboxylate ion is, thus lowering acidity	

Compound C is the least acidic as it has the electron donating group closer to the COO^- as compared to the 5 carbons arranged in a linear fashion.

25	An ester, $\text{C}_4\text{H}_8\text{O}_2$, undergoes the following reactions: • Reacts with hot sulfuric acid to give carboxylic acid Q and alcohol R. • The product from the oxidation of alcohol R gives a brick red precipitate with hot alkaline copper (II) complex. • Carboxylic acid Q reacts with lithium aluminium hydride in dry ether to give R. Which of the following could be the structure of the ester?	
	A	$\text{HCOOCH}(\text{CH}_3)_2$
	B	$\text{CH}_3\text{COOCH}_2\text{CH}_3$
	C	$\text{HCOOCH}_2\text{CH}_2\text{CH}_3$
	D	$\text{CH}_3\text{OCOCH}_2\text{CH}_3$
	Answer: B Ester in option B is made up of CH_3COOH and $\text{CH}_3\text{CH}_2\text{OH}$. CH_3COOH can be reduced to $\text{CH}_3\text{CH}_2\text{OH}$. The alcohol $\text{CH}_3\text{CH}_2\text{OH}$ can be oxidised to form aldehyde (CH_3CHO) which undergoes oxidation with Fehling's reagent to give brick-red ppt.	

Section B

For **questions 26-30**, one or more of the numbered statements **1** to **3** may be correct. Decide whether each of the statements is or is not correct. The responses **A** to **D** should be selected on the basis of:

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

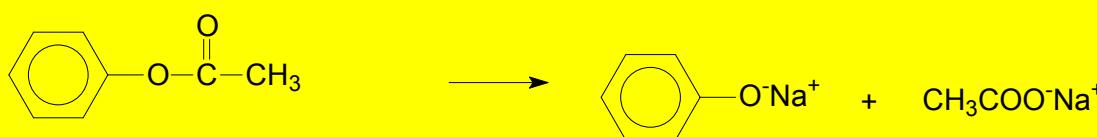
No other combination of statements is to be used as a correct response.

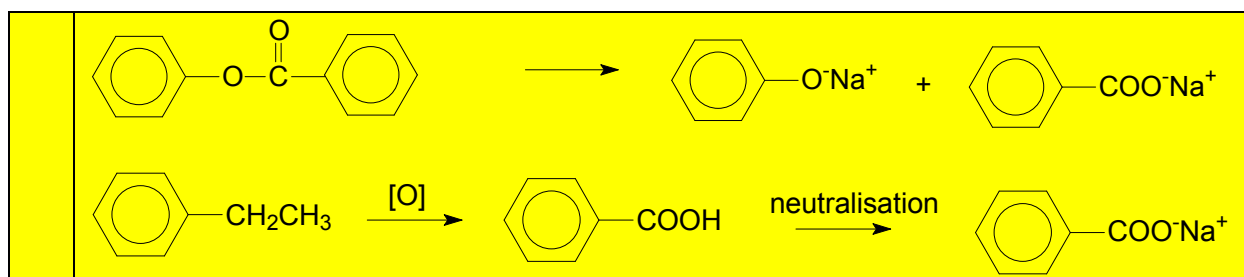
26	Which one of the following statements is true ?	
	1	AlCl_3 is covalent while AlF_3 is ionic.
	2	The melting point of Na is lower than that of Mg.
	3	An aqueous solution of HBr conducts electricity because HBr is ionic.
Answer: B HBr (aq) conducts electricity because it dissociates in water to form H^+ and Br^- ions that will conduct electricity. But HBr is a covalent compound.		

27	Which of the following mixture will result in an acidic buffer solution?	
	1	25 cm ³ of $\text{CH}_3\text{COO}^-\text{Na}^+$ and 25 cm ³ of CH_3COOH of same concentration
	2	25 cm ³ of $\text{CH}_3\text{COO}^-\text{Na}^+$ and 12.5 cm ³ of HCl of same concentration
	3	25 cm ³ of NaOH and 12.5 cm ³ of CH_3COOH of same concentration
Answer: B (1 and 2) Acidic buffer contains – weak acid and salt of conjugate base 1: CH_3COOH (weak acid) and CH_3COO^- (conjugate base) 2: $\text{CH}_3\text{COO}^- + \text{HCl} \rightarrow \text{CH}_3\text{COOH} + \text{Cl}^-$, since CH_3COO^- is in excess, the solution will contain both CH_3COO^- and CH_3COOH 3: NaOH is in excess, hence, the solution contains NaOH and CH_3COO^-		

28	Why is a solution of aluminium chloride more acidic than magnesium chloride?	
	1	The O-H bonds are weaker in $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ than in H_2O .
	2	Chloride ions react with water to form hydrochloric acid.
	3	Aluminium atom has high charge density.
Answer: D (1 only) Statement 1 is correct: The O-H bond is weakened due to the high charge density of Al^{3+} Statement 2 is wrong; the chloride ions do not react with water at all. Statement 3 is wrong; it should be aluminium ion has high charge density.		

29	Which of the following statements are true about the chemical reactions of cyclohexane, cyclohexene and methylbenzene?	
	1	One mole each of cyclohexene and methylbenzene will react with one mole and three moles of hydrogen gas respectively when heated in the presence of a catalyst under appropriate condition.
	2	All three compounds decolourise bromine gas under <i>uv</i> light.
	3	All three compounds decolourise hot acidified manganate (VII).
Answer: B (1 and 2)		

30	Which starting materials will give sodium benzoate on heating with aqueous sodium hydroxide under reflux?	
	1	phenyl ethanoate
	2	phenyl benzoate
	3	ethylbenzene and sodium manganate (VII)
Answer: C (2 and 3) 		



END